

# Seng Chon Chang

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## EDUCATION

### The Pennsylvania State University – Harrisburg

*Bachelor of Science in Electrical Engineering*

GPA: 4.0/4.0

Relevant Coursework: Calculus and Vector Analysis; Circuits and Devices; Cornerstone Engineering Design; Ordinary and Partial Differential Equations; Programming and Computation I: Fundamentals; Programming and Computation II: Data Structure

Middletown, PA

Aug 2025 - May 2026

### Northeastern University

*Bachelor of Science in Electrical Engineering*

Boston, MA

Sep 2026 - May 2029

## WORK

### Macao One Zero Technology Ltd., [facebook.com/10computerMacau](https://facebook.com/10computerMacau)

*IT Support Technician & System Solutions Assistant*, (4 hrs./wk., 2 wks./mo.)

Macao Island, Macao

Aug 2022 - Jun 2025

- Completed 180+ laptop/PC repairs; troubleshoot, part replacement, test; customer service & education
- Restored 4 TB+ files from 20+ corrupted hard drives; built 22+ custom PCs
- Co-operated weekly service livestream, Macao Baptist Church, on Ubuntu using OBS (scene switching, audio leveling, encoder/stream), Jan 2024

### Fuyue Xianju Vegetarian Restaurant, [bit.ly/d-vegan](https://bit.ly/d-vegan)

*Waiter*, (40 hrs./wk., 4 wks./mo.)

Macao Island, Macao

Summers 2020/2021

- Took orders, served meals, cleared plates and utensils, reset tables, organized dishware, cleaned dining area

## EXTRACURRICULAR PROJECTS

### Interactive Tools for Teaching Forward and Inverse Kinematics, [k.sengchon.com](https://k.sengchon.com)

Dec 2025 – July 2026

- Develop an online, interactive visual demo and program a physical 3 DoF robotic arm for in-class (ME 456 Introduction to Robotics) demonstration by Dr. Amit Banerjee, Associate Professor, Mechanical Engineering, Penn State Harrisburg

### A Bio-Inspired, Flexible, and Retractable Robotic Tail Exoskeleton, [isef.net/project/192569](https://isef.net/project/192569)

Aug 2024 – May 2025

- Founder/Captain/Team of 2; created a wearable robotic tail that reduces 70% gravitational torque, which negatively correlates with human gait stability
- Drew Inspiration from cats' tails; reviewed literature; designed 83+ iterations in Fusion 360; printed 12+ 3D prototypes; programmed C++ library for BLDC motors; deployed ROS2 on JetsonNano for modularity; integrated nonlinear IK solver for soft robotics; developed a real-time balance algorithm; co-designed evaluation methods

### NeRF Technology: An Economic Approach to 3D Scanning, [isef.net/project/16240](https://isef.net/project/16240)

July 2023 – May 2024

- Member/Team of 3; created a high-quality 3D scanner capable of digitizing translucent museum artifact
- Innovated a pre-processing workflow in Neural Radiance Fields (NeRF); engineered a 3-DoF robotic arm and a stepper-motor rotary table in Fusion 360 (161 iterations); designed & conducted evaluations

### A Line-Following, Lightweight, Two-Wheeled Robot, [sengchon.com/lbot](https://sengchon.com/lbot)

Sep 2023 – Nov 2023

- Founder/Captain/Team of 2; designed from scratch a line-following robot with a carbon-fiber chassis for faster movement
- 3D-printed herringbone gearbox with 4.55:1 reduction; implemented power regulation with LM2596 buck (12V-5V); used 2 CD4051B multiplexers to read 15 analog sensors; established PID controller; designed board-to-board protocol; developed a web UI (React + Material UI) with features for connection, driving, PID tuning, calibration

### Additional Independent/Individual Maker Projects, [sengchon.com](https://sengchon.com)

Since 2018

- Brushless DC Motor Driver Based on F.O.C.
- CoreXY-based Drawing Robot
- Rubik's Cube Solving Robot
- Two-legged Servo Robot

- User-Generated Business Review Map Web App

## HONORS AND AWARDS

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- Honorific Title, Certificates of Merit, team of 2, *A Robotic Tail Exoskeleton*, Government of the Macao Special Administrative Region (MSAR) Jan 2026
- 3rd, \$1200, Grand Award, team of 2, *A Robotic Tail Exoskeleton*, Engineering Technology: Statics and Dynamics category, Regeneron ISEF May 2025
- Finalist, team of 3, *NeRF Technology: An Economic Approach to 3D Scanning*, Robotics and Intelligent Machines category, Regeneron ISEF May 2024
- 2x 1st place, Elite Talent & Innovation Competition, Education and Youth Development Bureau, MSAR Nov 2024/2023
- 2nd/3rd places, Selection Competition, Macao Youth for Integrated Robotics Science Activity May 2022/2021
- 2nd place, High School Regular, Macao Regional, World Robot Olympiad July 2021

## CAMPUS & COMMUNITY INVOLVEMENT

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- Chair, Technical Committee**, *Penn State Harrisburg IEEE Student Branch* (8 hrs./wk., 6 mos./yr.) Middletown, PA
- Mentor/Lead the technical team of 19 to organize 16 1-2 hour-long technical workshops each semester Sep 2025 - June 2026
  - Organize/Host/Teach 5 technical workshops
    - Past events: Design & 3D-Print a Phone Holder; 3D-Printing Crash Course; Halloween Soldering Workshop; Design & 3D-Print Your Own Key Chain with Fusion 360!; 4-40+ student participants; [bit.ly/d-wshop](https://bit.ly/d-wshop)
  - Initiated request and received equipment sponsorship (including an A1 3D printer and AMS - Automatic Material System), Bambu Lab, January 2026

- Founder/President**, *Design and Applied Technology club, Instituto Salesiano High School* Macao Island, Macao  
(7 hrs./mo.) July 2024 - May 2025
- Promoted STEM interests on campus through technology, programming, design, and innovation
  - Recruited 60+ members; established competition, marketing, and operations divisions; hosted weekly officer meetings
  - Organized special events: school open-day club showcase (300+ students), Feb 2025; fundraiser booth (700+ public) at the 55th Macao Charity Bazaar, Nov 2024

## SKILLS

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### Technical:

|                                                   |                                    |                                |
|---------------------------------------------------|------------------------------------|--------------------------------|
| Embedded Systems Design                           | PCB Design                         | SMD Soldering                  |
| Motor Control (DC: H-bridge; BLDC: F.O.C.)        | Power Regulation (DC-DC)           | ROS2/micro-ROS Integration     |
| Programming (C++, Python, Java, JavaScript)       | 3D CAD (Fusion 360; AutoCAD)       | PID Control                    |
| Prototyping: (FDM/SLA 3D printing, laser cutting) | Forward/Inverse Kinematics Solving | Version Control (Git)          |
| Device Repair/Troubleshooting                     | Frontend/Backend Development       | Computer Vision Implementation |

### Laboratory:

|                                           |                                     |                         |
|-------------------------------------------|-------------------------------------|-------------------------|
| Oscilloscope/Function Generator Operation | Breadboarding/Perfboard Prototyping | Circuit Troubleshooting |
| Multimeter (DMM) Measurement              | Soldering/Rework (TH/SMD)           | Serial Debugging        |

### Language:

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|-----------------------------|--------------------|-------------------|
| English (Full professional) | Cantonese (Native) | Mandarin (Native) |
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